CIVL6410 Transport Network Modelling

Week 6:

Traffic Assignment Problems

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Outline

- **Traffic Assignment Algorithms**
  - All-of-Nothing (AON) assignment
  - Convergence Criteria
  - Link-based Algorithms
    - Method of Successive Averages (MSA)

- References
  - Bolyes, Lownes, and Unnikrishnan (BLU) Book
    - Chapter 5. The Traffic Assignment Problem
    - Chapter 6. Algorithms for Traffic Assignment

Introduction to Assignment Algorithms

Traffic Assignment Problem

- Given:
  - A graph representation of the urban transportation network
  - The associated link performance functions
  - An origin-destination (OD) matrix

- **Find the flow (and the travel time) on each of the network links.**

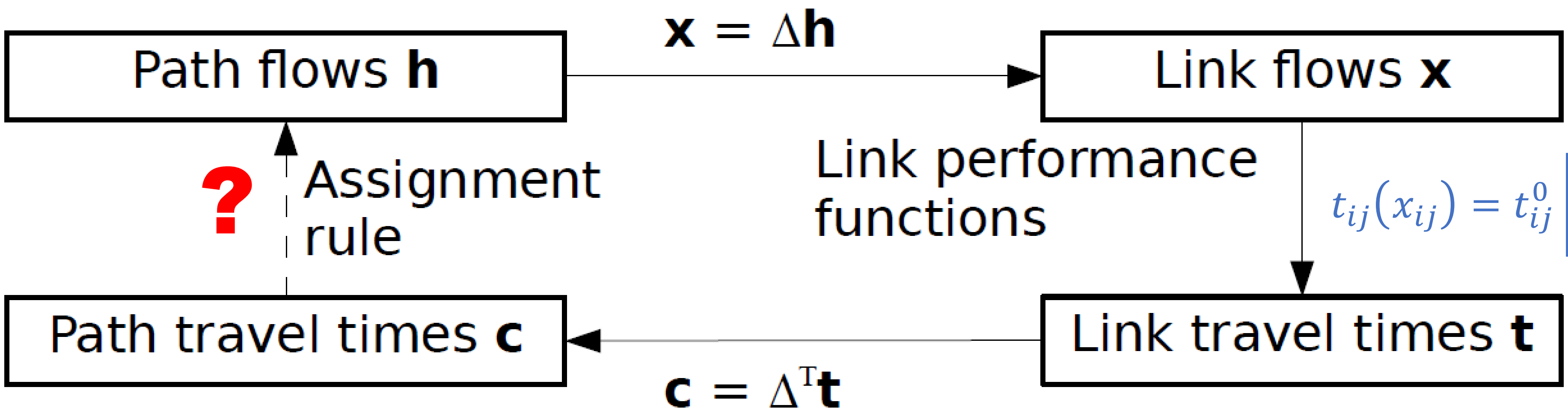
Mathematical Notation (following the BLU book)

- N : the set of nodes, i.e., node $i \in N$
- A : the set of links, i.e., link $(i, j) \in A$
- Z : the set of zone centroids, which is a subset of node set N (i.e., $Z \subseteq N$)
  - r, s: the origin and destination zone centroids, respectively; $r \in Z$ and $s \in Z$
  - (r, s) : the OD-pair between origin r and destination s; $(r, s) \in Z^2$
- Π^{rs}: the set of paths between origin r and destination s, i.e., path $\pi \in \Pi^{rs}$
- Π : the set of all paths, i.e., $\bigcup_{(r,s)\in Z^2} \Pi^{rs}$ (i.e., $\Pi^{rs} \subseteq \Pi$)
- d^{rs} : trip demand between origin r and destination s; $\mathbf{d} = (\dots, d^{rs}, \dots)$
- x_{ij} : link flow on link (i, j); $\mathbf{x} = \left(\dots, x_{ij}, \dots\right)$
- t_{ij} : link travel time on link (i, j); $\mathbf{t} = \left(\dots, t_{ij}, \dots\right)$
- h^{π}: path flow on path π; $\mathbf{h} = (\dots, h^{\pi}, \dots)$
- c^{π}: path travel time on path π; $\mathbf{c} = (\dots, c^{\pi}, \dots)$
- δ_{ij}^{π}: indicator variable; $\delta_{ij}^{\pi} = 1$ if link (i, j) is on path π and $\delta_{ij}^{\pi} = 0$ otherwise
- $\boldsymbol{\Delta}$: *link-path adjacency matrix* with δ_{ij}^{π} for link row (i, j) and path column π

Traffic Assignment Problem

- **What assignment rule?** How each driver chooses his/her route, given a set of routes and their predicted travel times?

$$x_{ij} = \sum_{r\in Z}\sum_{s\in Z}\sum_{\pi\in\Pi^{rs}} \delta_{ij}^{\pi} h^{\pi}$$



$$t_{ij}(x_{ij}) = t_{ij}^{0}\left[1 + \alpha\left(\frac{x_{ij}}{u_{ij}}\right)^{\beta}\right]$$

$$c^{\pi} = \sum_{(i,j)\in A} \delta_{ij}^{\pi} t_{ij}$$

Figure 4.2: Traffic assignment as an iterative process.

User Equilibrium (UE) Assignment

- At the **User-Equilibrium** (UE), '*all **used** paths between the same origin and destination have **equal** and **minimal** travel times*'.

- Assumptions
  - **Shortest Path** assumption**:** *Each driver wants to choose the path between their origin and their destination with the least travel time*.
  - **Perfect Information** assumption: *Drivers have perfect knowledge of link travel times*.

Overall Steps of Equilibrium Solution Algorithm

- Broadly speaking, all **equilibrium solution algorithms** repeat the following three steps:
  1. Find the shortest path between each origin and each destination.
  2. Shift travellers from slower paths to faster ones.
  3. Recalculate link flows and travel times after the shift, and return to step 1 unless we are close enough to equilibrium.

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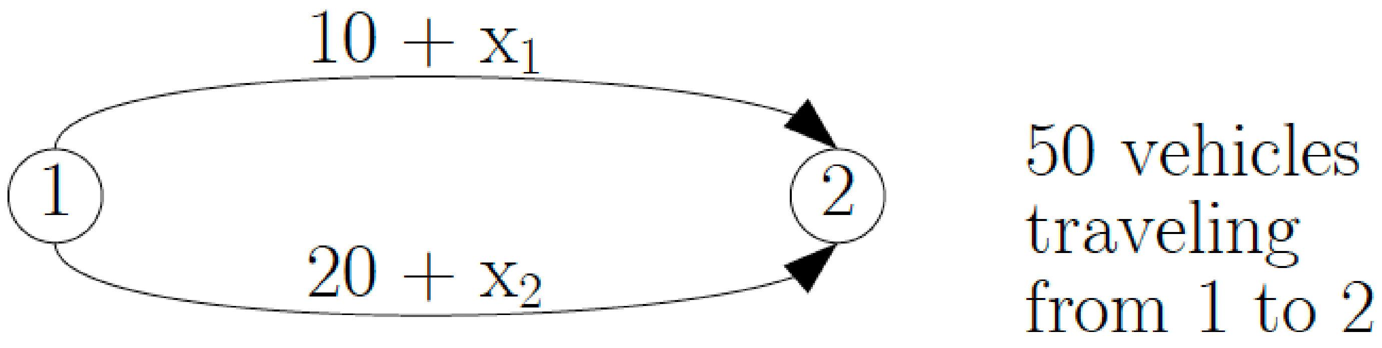
Step 2 requires the most care.
If we shift too few, it will take a long time to get to the equilibrium solution.
If we shift too many, it may not converge at all as the flow keeps flip-flopping between paths.

All-or-Nothing (AON) Assignment

- The '***all-or-nothing***' assignment is an assignment rule that loads **all of the demand** for a given OD-pair onto the shortest path for that OD.
- Usually this is the case of 'shifting too many'.

- A solution algorithm using ***all-or-nothing*** assignments
  - **Step 0:** *Initialisation.* Find the shortest paths based on the free-flow travel times (assuming no travellers on the network). Perform *all-or-nothing* assignment (i.e., assign all OD demand to the shortest path of each OD). Obtain an initial link flow and the resulting link travel times.
  - **Step 1:** *Assignment.* Find the shortest paths based on the new link travel times. Perform *all-or-nothing* assignment.
  - **Step 2:** *Update.* Calculate the link flows and travel times after the shift.
  - **Step 3:** *Convergence test.* If the convergence criterion is satisfied, stop; otherwise return to Step 1.

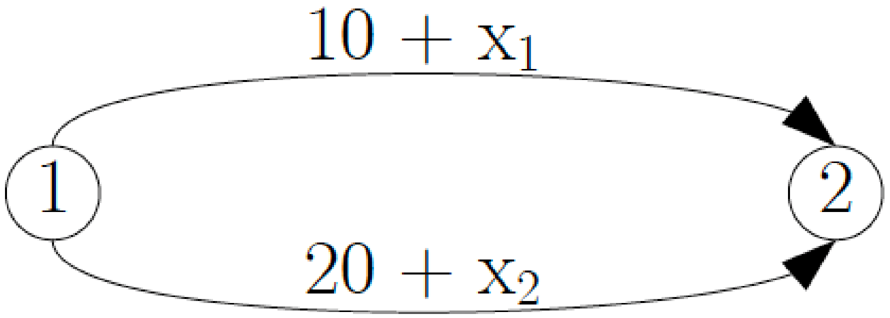
Example



[BLU Book] Figure 6.1: Two-link example for demonstration.

- UE solution by analytical equations:
  - $10 + x_1 = 20 + x_2$ and $x_1 + x_2 = 50 \rightarrow x_1 = 30$ and $x_2 = 20$

Shifting Too Many

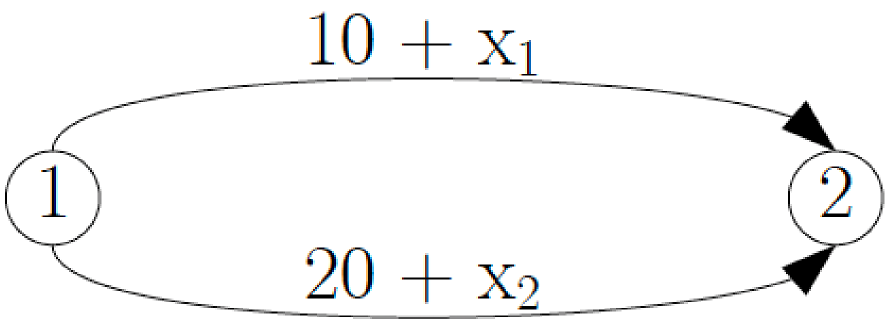


50 vehicles traveling from 1 to 2

Iteration Number	Algorithmic Step	Link	
		1 (top link)	2 (bottom link)
0	- Initialisation	$t_1 = 10$	$t_2 = 20$
	- Initial link flow and travel time	$x_1 = \mathbf{50}$ $t_1 = 60$	$x_2 = 0$ $t_2 = 20$
1	- All-or-nothing - Update time	$x_1 = 0$ $t_1 = 10$	$x_2 = \mathbf{50}$ $t_2 = 70$
2	- All-or-nothing - Update time	$x_1 = \mathbf{50}$ $t_1 = 60$	$x_2 = 0$ $t_2 = 20$
3	- All-or-nothing - Update time	$x_1 = 0$ $t_1 = 10$	$x_2 = \mathbf{50}$ $t_2 = 70$
⋮	⋮	⋮	⋮

If we kept up this process, we'd keep bouncing back and forth between two solutions and never reach the equilibrium with $x_1 = 30$ and $x_2 = 20$

Shifting Too Few



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Iteration Number	Algorithmic Step	Link	
		1 (top link)	2 (bottom link)
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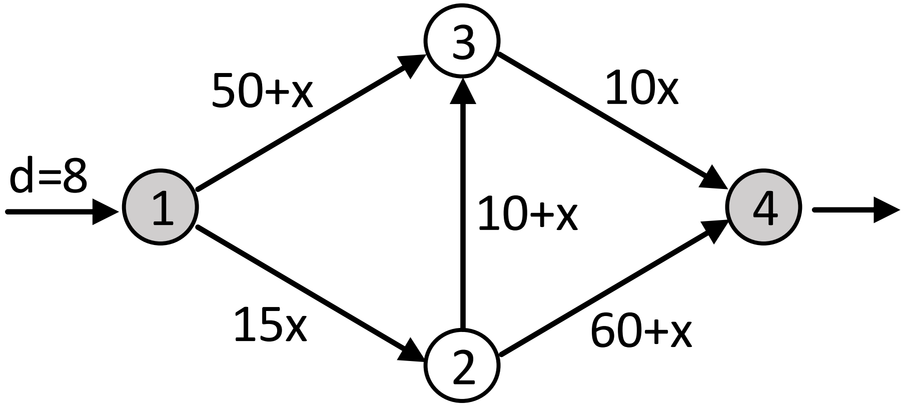
	- Initial link flow and travel time	$x_1 = \mathbf{50}$ $t_1 = 60$	$x_2 = \mathbf{0}$ $t_2 = 20$
1	- One-at-a-time - Update time	$x_1 = \mathbf{49}$ $t_1 = 59$	$x_2 = \mathbf{1}$ $t_2 = 21$
2	- One-at-a-time - Update time	$x_1 = \mathbf{48}$ $t_1 = 58$	$x_2 = \mathbf{2}$ $t_2 = 22$
3	- One-at-a-time - Update time	$x_1 = \mathbf{47}$ $t_1 = 57$	$x_2 = \mathbf{3}$ $t_2 = 23$
⋮	⋮	⋮	⋮

If we kept up this process, it would take long time to reach the equilibrium with $x_1 = 30$ and $x_2 = 20$

Exercise

Consider the network on the right with one OD-pair from node 1 to node 4 and OD demand of 8 vehicles.

- Find the All-or-Nothing (AON) assignment link flows.
  - Paths: $\pi_1 = [1,2,4],\ \pi_2 = [1,3,4],\ \pi_3 = [1,2,3,4]$



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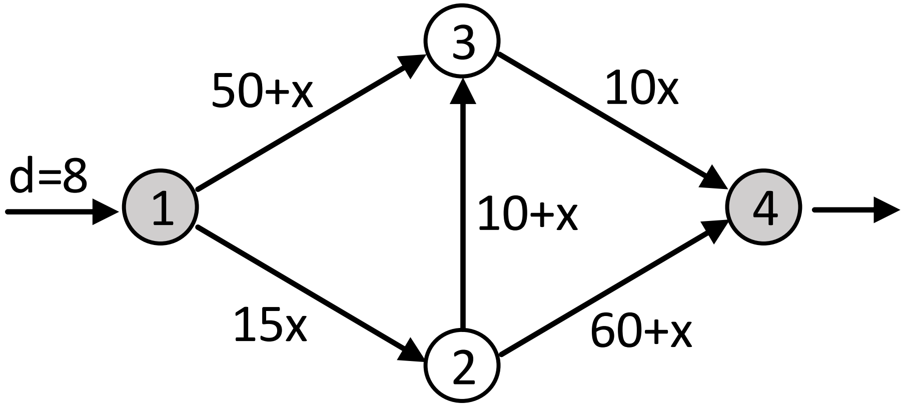

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    - Initial link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [\qquad\qquad\qquad]$
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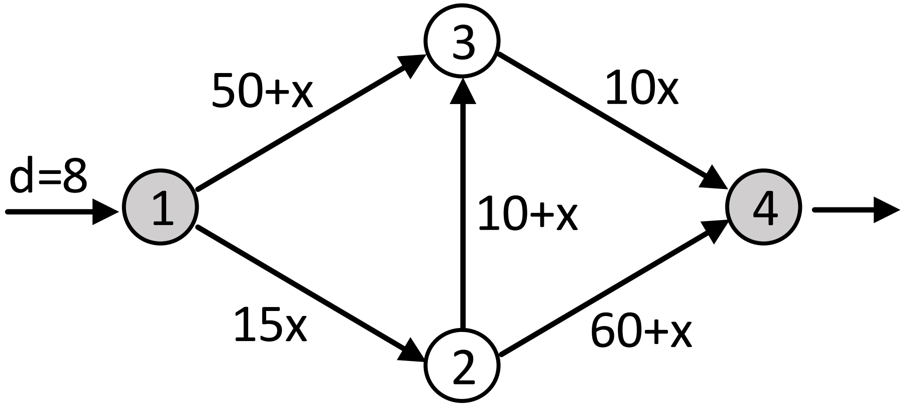

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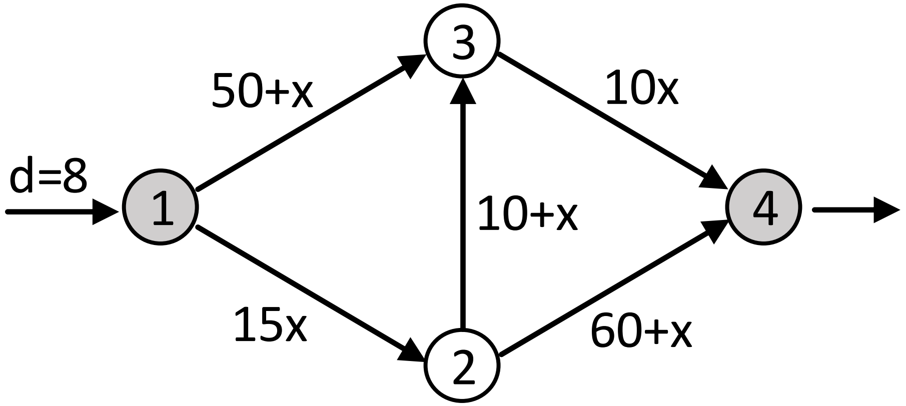

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    - Cost of path $\pi_1 : c^1 = \ ?$
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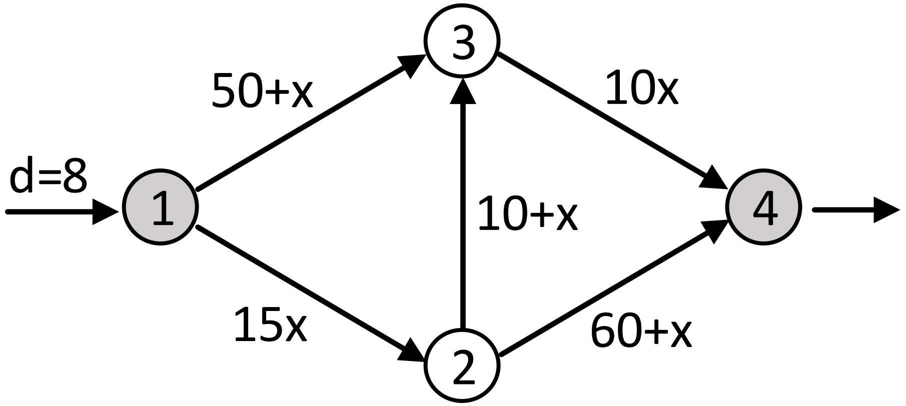

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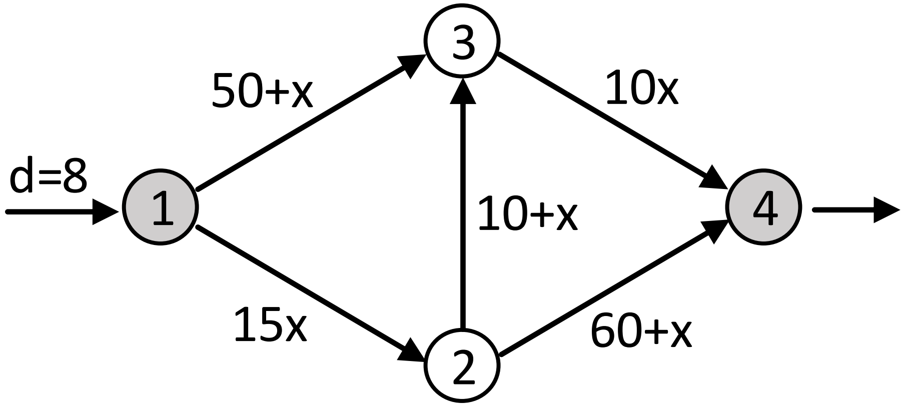

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  - Assign all OD demand to the shortest path and update the link flows.
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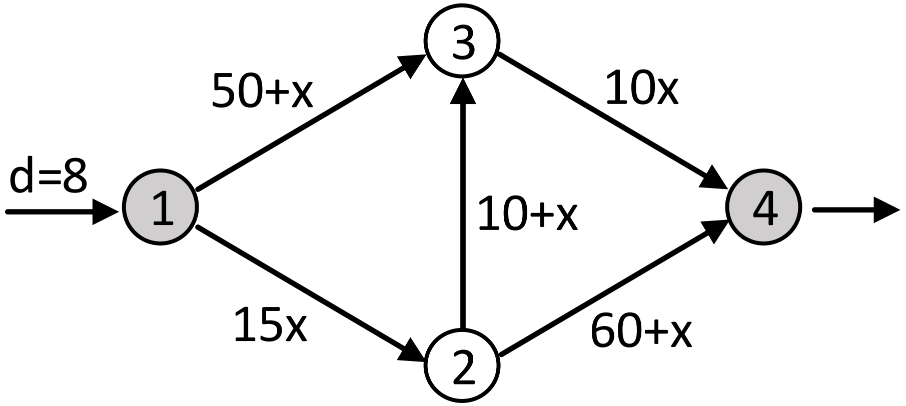

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  - Assign all OD demand to the shortest path and update the link flows.
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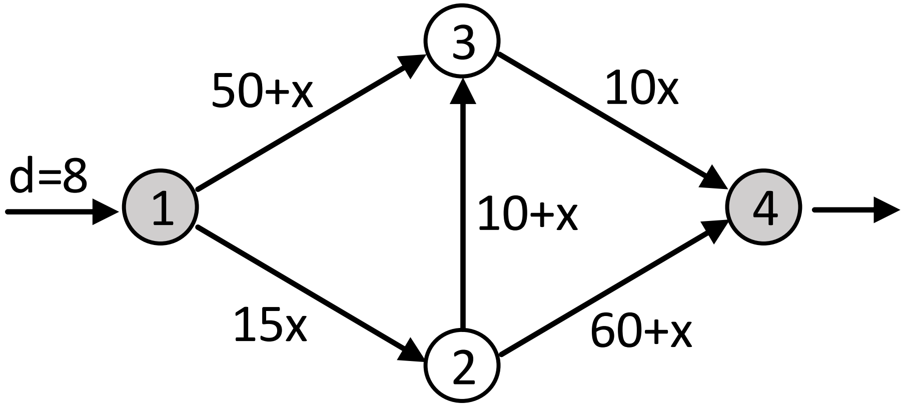

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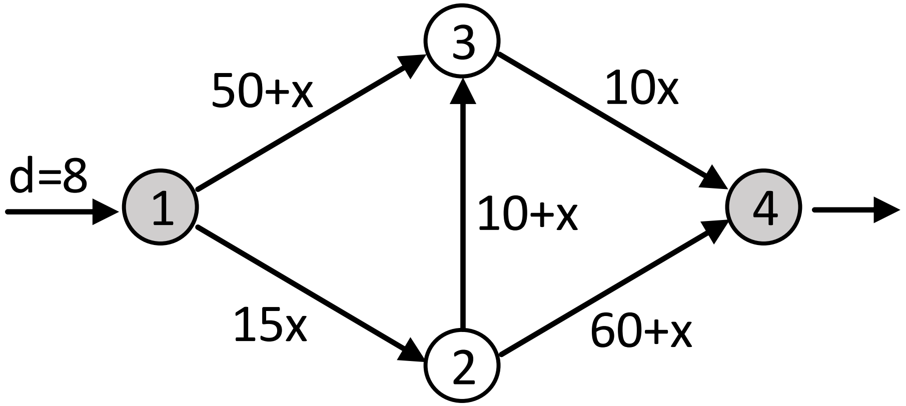

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    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 120 + 60 = 180$
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 50 + 80 = 130$ ← **SHORTEST PATH**
    - Cost of path $\pi_3 : c^3 = t_{12} + t_{23} + t_{34} = 120 + 18 + 80 = 218$

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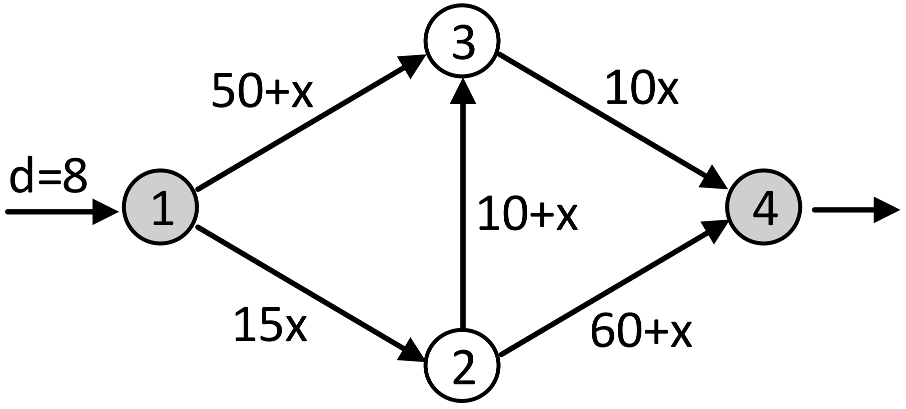

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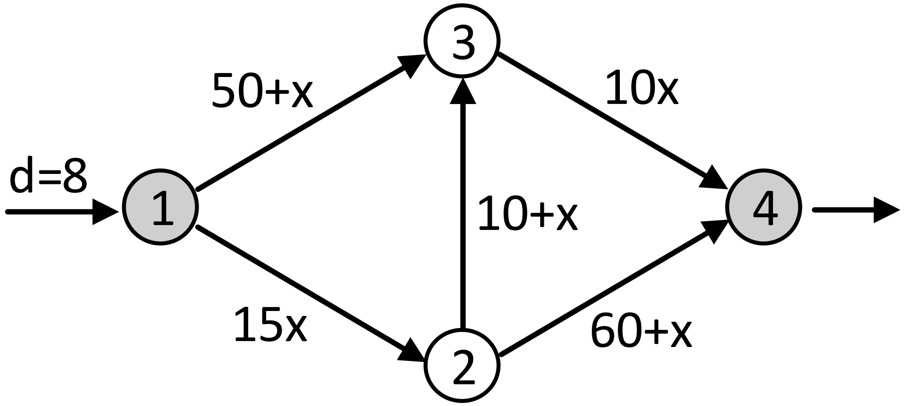

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This process can continue to Iterations 2, 3, 4, etc. but when should we stop?

Convergence Criteria

Convergence Criteria

- A general issue is how one chooses to stop the iterative process, that is, how one knows when a solution is '**good enough**' or close enough to **equilibrium**.
- This is called a ***convergence criterion***.
- Usually, some **measure of convergence** (γ) is defined, which approaches zero as the solution converges, and checked against some small number (e.g., $\gamma < \epsilon$).

- Common convergence criteria:
  - **Relative Gap (RG)**
  - **Average Excess Cost (AEC)**

Total Travel Time Measures: Actual vs. Ideal

Total System Travel Time (TSTT)

- $TSTT = \sum_{\pi \in \Pi} c^{\pi} h^{\pi} = \sum_{(i,j) \in A} t_{ij} x_{ij}$
  - t_{ij}, x_{ij}: link travel time and link flow on link (i, j)
  - c^{π}, h^{π}: path travel time and path flow on path π
- **Sum of all travellers' experienced travel times** (given current link travel times)
- What travellers *actually* experience

Shortest Path Travel Time (SPTT)

- $SPTT = \sum_{(r,s) \in Z^2} \kappa^{rs} d^{rs}$
  - κ^{rs} : shortest path travel time between origin r and destination s
  - d^{rs}: demand for OD-pair (r, s)
- **The ideal total travel time if every traveller used their shortest path** (given current link travel times).
- The *best achievable* total system travel time under current conditions.

$$\boxed{TSTT \geq SPTT}$$

Actual Ideal

Convergence Criteria: **Relative Gap**

- **Relative Gap (RG):** $$RG = \frac{\sum_{(i,j)\in A} t_{ij}x_{ij}}{\sum_{(r,s)\in Z^2} \kappa^{rs} d^{rs}} - 1 = \frac{\mathbf{t}\cdot\mathbf{x}}{\boldsymbol{\kappa}\cdot\mathbf{d}} - 1 = \frac{TSTT}{SPTT} - 1$$

  - κ^{rs} : the shortest travel time spent on the fastest path between origin r and destination s
  - TSTT: the total system travel time (**Actual**)
  - SPTT: shortest path travel time (**Ideal**)

- The relative gap represents **the relative difference between the total system travel time based on travellers' *actual* paths and the total system travel time based on the *shortest* paths** ($RG = \frac{\mathbf{t}\cdot\mathbf{x}-\boldsymbol{\kappa}\cdot\mathbf{d}}{\boldsymbol{\kappa}\cdot\mathbf{d}}$).
- The relative gap is always non-negative, and it is equal to zero if and only if the flows $\mathbf{x}$ satisfy the principle of user equilibrium.
- For most practical purposes, $RG < 10^{-4}\sim 10^{-6}$ is small enough.
- Drawback: it is unitless and does not have an intuitive meaning

Convergence Criteria: **Average Excess Cost**

- **Average Excess Cost (AEC):**

$$AEC = \frac{\sum_{(i,j)\in A} t_{ij}x_{ij} - \sum_{(r,s)\in Z^2} \kappa^{rs} d^{rs}}{\sum_{(r,s)\in Z^2} d^{rs}} = \frac{\mathbf{t}\cdot\mathbf{x} - \boldsymbol{\kappa}\cdot\mathbf{d}}{\mathbf{d}\cdot\mathbf{1}} = \frac{TSTT - SPTT}{\text{Total Demand}}$$

  - AEC represents **the average difference between the travel time on each traveller's *actual* path and the travel time on the *shortest* path available to the traveller**.
  - AEC = **Actual travel time** – **Ideal travel time** (shortest path)
  - Unlike the relative gap (RG), AEC has **units of time**, and is thus easier to interpret.

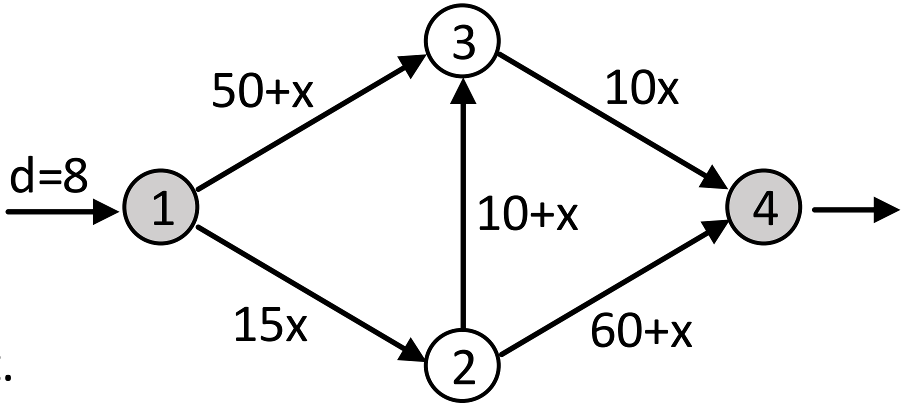
Additional Measure: **Average Shortest Path Travel Time**

Average Shortest Path Travel Time (AvgSPTT):

$$AvgSPTT = \frac{\sum_{(r,s)\in Z^2} \kappa^{rs} d^{rs}}{\sum_{(r,s)\in Z^2} d^{rs}} = \frac{\boldsymbol{\kappa} \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{1}} = \frac{SPTT}{\text{Total Demand}}$$

- AvgSPTT represents the **average shortest path travel time per traveller** that travellers *could* achieve if everyone used the shortest paths available to them (under the given traffic assignment).
- AEC = $\frac{TSTT}{\mathbf{d}\cdot\mathbf{1}} - \frac{SPTT}{\mathbf{d}\cdot\mathbf{1}}$ = Actual – AvgSPTT (Ideal baseline)
- Interpretation:
  - Low AvgSPTT → network is inherently fast
  - High AvgSPTT → even the best routes are slow (congestion or long distances)

Exercise



Consider the network on the right with one OD-pair from node 1 to node 4 and OD demand of 8 vehicles.

- Check convergence after one All-or-Nothing (AON) assignment.
  - Paths: $\pi_1 = [1,2,4],\ \pi_2 = [1,3,4],\ \pi_3 = [1,2,3,4]$

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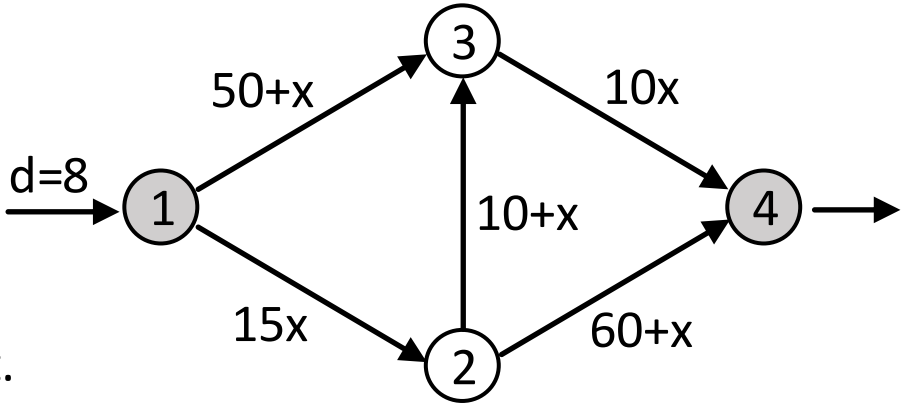

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  - Find the shortest path: calculate path cost for all three paths and find the minimum cost path.
    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 0 + 60 = 60$
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 50 + 0 = 50$
    - Cost of path $\pi_3 : c^3 = t_{12} + t_{23} + t_{34} = 0 + 10 + 0 = 10$ ← **SHORTEST PATH**
  - Assign all OD demand to the shortest path and update the link flows.
    - AON path flows $\mathbf{h} = [h^1, h^2, h^3] = [0,0,8]$
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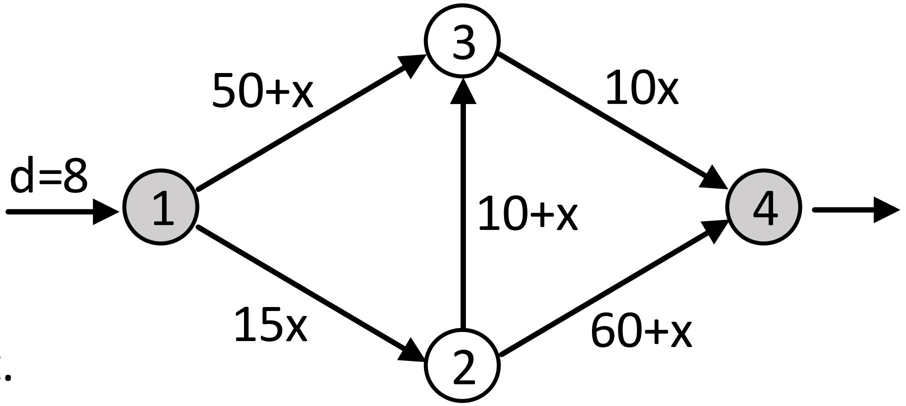

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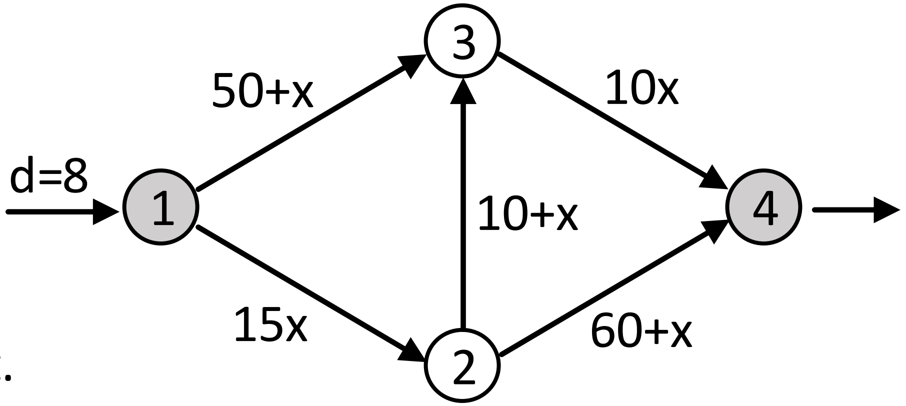

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$$RG = \frac{\sum_{(i,j)\in A} t_{ij}x_{ij}}{\sum_{(r,s)\in Z^2} \kappa^{rs}d^{rs}} - 1 = \frac{\mathbf{t}\cdot\mathbf{x}}{\boldsymbol{\kappa}\cdot\mathbf{d}} - 1$$

$$AEC = \frac{\sum_{(i,j)\in A} t_{ij}x_{ij} - \sum_{(r,s)\in Z^2} \kappa^{rs}d^{rs}}{\sum_{(r,s)\in Z^2} d^{rs}} = \frac{\mathbf{t}\cdot\mathbf{x} - \boldsymbol{\kappa}\cdot\mathbf{d}}{\mathbf{d}\cdot\mathbf{1}}$$

$\mathbf{t}\cdot\mathbf{x}$ =120(8)+50(0)+18(8)+60(0)+80(8)=1744

$\boldsymbol{\kappa}\cdot\mathbf{d}$ =130(8)=1040

$$RG = \frac{\mathbf{t}\cdot\mathbf{x}}{\boldsymbol{\kappa}\cdot\mathbf{d}} - 1 = \frac{1744}{1040} - 1 = 0.6769$$

$$AEC = \frac{\mathbf{t}\cdot\mathbf{x} - \boldsymbol{\kappa}\cdot\mathbf{d}}{\mathbf{d}\cdot\mathbf{1}} = \frac{1744-1040}{8} = 88$$

UE Assignment Algorithms

Types of UE Assignment Algorithms

- **Path-based algorithms**
  - Decision variables: path flows $\mathbf{h} = (\dots, h^{\pi}, \dots)$
  - These algorithms:
    - Iteratively shift path flows between paths to equalise travel times across all used paths for an OD pair
    - Directly adjust path choices and explicitly track the flow on each path
    - Can converge faster and provide a more direct interpretation of how travellers switch between routes
- **Link-based algorithms**
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We will use **link-based algorithms** as these are the simplest to understand and implement.

Mathematical Formulation for UE Assignment

- The goal is to find the link flows, $\mathbf{x} = (\dots, x_{ij}, \dots)$, that satisfy the user-equilibrium criterion when all the origin-destination demand, $\mathbf{d} = (\dots, d^{rs}, \dots)$, have been appropriately assigned.
- This link flow pattern can be obtained by solving the following **mathematical *optimisation* problem**:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \min_{\mathbf{x}} \left(\sum_{(i,j)\in A} \int_0^{x_{ij}} t_{ij}(x)dx \right)$$

s.t. (subject to)

$$x_{ij} = \sum_{\pi\in\Pi} h^\pi \delta_{ij}^\pi \quad \forall (i,j) \in A$$

$$d^{rs} = \sum_{\pi\in\Pi^{rs}} h^\pi \quad \forall (r,s) \in Z^2$$

$$h^\pi \geq 0 \quad \forall \pi \in \Pi$$

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Objective Function

Optimisation Problem
- 'min': Minimisation
- 'max': Maximisation

Decision Variables

$$\min_{\mathbf{x}} \boxed{f(\mathbf{x})} = \min_{\mathbf{x}} \left(\boxed{\sum_{(i,j)\in A} \int_0^{x_{ij}} t_{ij}(x)dx} \right)$$

← The objective function, $f(\mathbf{x})$, to be ***minimised*** w.r.t (with respect to) decision variable $\mathbf{x}$

s.t. (subject to)

Constraints

$$\boxed{\begin{aligned} x_{ij} &= \sum_{\pi\in\Pi} h^\pi \delta_{ij}^\pi && \forall (i,j) \in A \\ d^{rs} &= \sum_{\pi\in\Pi^{rs}} h^\pi && \forall (r,s) \in Z^2 \\ h^\pi &\geq 0 && \forall \pi \in \Pi \end{aligned}}$$

← A set of constraints

- *Link-path flow relationship*
- *Flow conservation* (the sum of path flows should be equal to the OD demand)
- *Nonnegativity condition* (the path flows should be non-negative)

***Beckmann's* Formulation** for UE Traffic Assignment

$$\min_{\mathbf{x}} f(\mathbf{x}) = \min_{\mathbf{x}} \left(\sum_{(i,j)\in A} \int_0^{x_{ij}} t_{ij}(x)dx \right)$$

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$$h^\pi \geq 0 \quad \forall \pi \in \Pi$$

- This formulation is known as ***Beckmann's* Formulation**, developed by Martin Beckmann in 1956.
- This formulation guarantees that the optimal solution $\mathbf{x}^*$ (i.e., link flows) satisfies the UE principle: "all used paths between an origin-destination pair have equal and minimal travel costs".

***Beckmann's* Formulation** for UE Traffic Assignment

- The objective function, $f(\mathbf{x}) = \sum_{(i,j)\in A} \int_0^{x_{ij}} t_{ij}(x)dx$, does not have any intuitive economic or behavioural interpretation. This is a purely mathematical construct, whose solution satisfies the UE conditions.

$$\min_{\mathbf{x}} f(\mathbf{x}) = \min_{\mathbf{x}} \left(\sum_{(i,j)\in A} \int_0^{x_{ij}} t_{ij}(x)dx \right)$$

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- **How Is It Used in Traffic Assignment Algorithms?**
  - **Backmann's function** is used to guide the iterative update of link flows within a link-based algorithm like *Frank-Wolfe* algorithm
  - At each iteration, the amount of link flow shifts is determined using a '**step size**' parameter.
    - The **step size** is a numerical factor that controls the proportion of flow moved from the current solution toward the newly computed direction.
  - This **step size** is chosen to minimise Beckmann's objective function toward its optimal (minimum) value, which corresponds to the **User Equilibrium** state.

Overview of Link-based Algorithms

- General framework:

  1. Generate an initial solution $\mathbf{x}$ (current link flows).
  2. Generate a target solution $\mathbf{x}^*$ (new link flows).
  3. Update the current solution: $\boxed{\mathbf{x} \leftarrow \lambda \mathbf{x}^* + (1-\lambda)\mathbf{x}}$ for some $\lambda \in [0,1]$.
  4. Calculate the new link travel times.
  5. If the convergence criterion is satisfied, stop; otherwise return to step 2.

Step size λ determines how much to shift.

- Two solution algorithms
  - **Method of Successive Averages (MSA)**: uses a *fixed* step size
  - **Frank-Wolfe (FW)**: uses an *adaptive* step size

Method of Successive Averages (MSA)

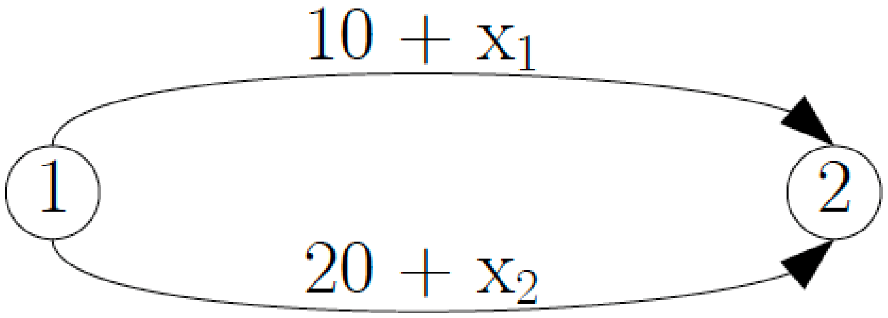
$$\boxed{\mathbf{x} \leftarrow \lambda \mathbf{x}^* + (1-\lambda)\mathbf{x}}$$

- The initial $\mathbf{x}$ is the current set of link flows.
- The target $\mathbf{x}^*$ is link flows obtained from the ***all-or-nothing*** assignment.
- In the i^{th} iteration, MSA step size is:

$$\boxed{\lambda = \frac{1}{(i+1)}} \Rightarrow \mathbf{x} \leftarrow \frac{1}{(i+1)}\mathbf{x}^* + \left(1 - \frac{1}{(i+1)}\right)\mathbf{x}$$

- $i = 0, \lambda = 1 \rightarrow i = 1, \lambda = \frac{1}{2} \rightarrow i = 2, \lambda = \frac{1}{3} \rightarrow i = 3, \lambda = \frac{1}{4} \rightarrow \cdots$
- At first, we take big steps, so hopefully we get somewhere close to equilibrium quickly.
- Eventually, we only move small amounts of flow, so there is no worry of flip-flopping.

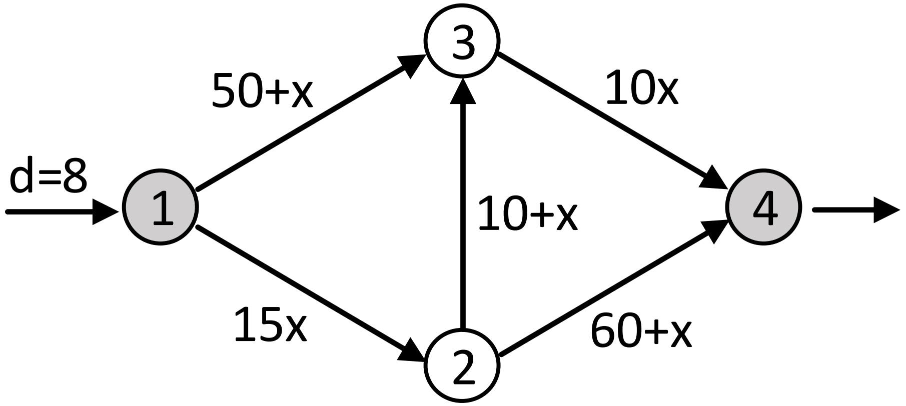
MSA Example



50 vehicles traveling from 1 to 2

Iter. No.	Algorithmic Step	Link		$\mathbf{x} \leftarrow \lambda \mathbf{x}^* + (1-\lambda)\mathbf{x}$	Convergence Test ($RG = \frac{\mathbf{t}\cdot\mathbf{x}}{\boldsymbol{\kappa}\cdot\mathbf{d}} - 1$)
		1 (top link)	2 (bottom link)		
0	- Initialisation	$t_1 = 10$	$t_2 = 20$		
	- Initial link flow and travel time	$x_1 = 50$ $t_1 = 60$	$x_2 = 0$ $t_2 = 20$		$RG = \frac{60\times50+20\times0}{20\times50} - 1 = 2$
1	- MSA - Update time	$x_1 = 25$ $t_1 = 35$	$x_2 = 25$ $t_2 = 45$	$x_1 \leftarrow (1/2)(0) + (1/2)(50) = 25$ $x_2 \leftarrow (1/2)(50) + (1/2)(0) = 25$	$RG = \frac{35\times25+45\times25}{35\times50} - 1 = 0.143$
2	- MSA - Update time	$x_1 = 33.33$ $t_1 = 43.33$	$x_2 = 16.67$ $t_2 = 36.67$	$x_1 \leftarrow (1/3)(50) + (2/3)(25) = 33.33$ $x_2 \leftarrow (1/3)(0) + (2/3)(25) = 16.67$	$RG = \frac{43.33\times33.33+36.67\times16.67}{36.67\times50} - 1 =$ 0.121
3	- MSA - Update time	$x_1 = 25$ $t_1 = 35$	$x_2 = 25$ $t_2 = 45$	$x_1 \leftarrow (1/4)(0) + (3/4)(33.33) = 25$ $x_2 \leftarrow (1/4)(50) + (3/4)(16.67) = 25$	$RG = \frac{35\times25+45\times25}{35\times50} - 1 = 0.143$
4	- MSA - Update time	$x_1 = 30$ $t_1 = 40$	$x_2 = 20$ $t_2 = 40$	$x_1 \leftarrow (1/5)(50) + (4/5)(25) = 30$ $x_2 \leftarrow (1/5)(0) + (4/5)(25) = 20$	$RG = \frac{40\times30+40\times20}{40\times50} - 1 = 0$ (stop)

Exercise



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  - Perform an AON assignment to obtain the initial ‘current’ solution.
    - AON path flows $\mathbf{h} = [h^1, h^2, h^3] = [0,0,8]$
    - The resulting AON link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [8,0,8,0,8]$
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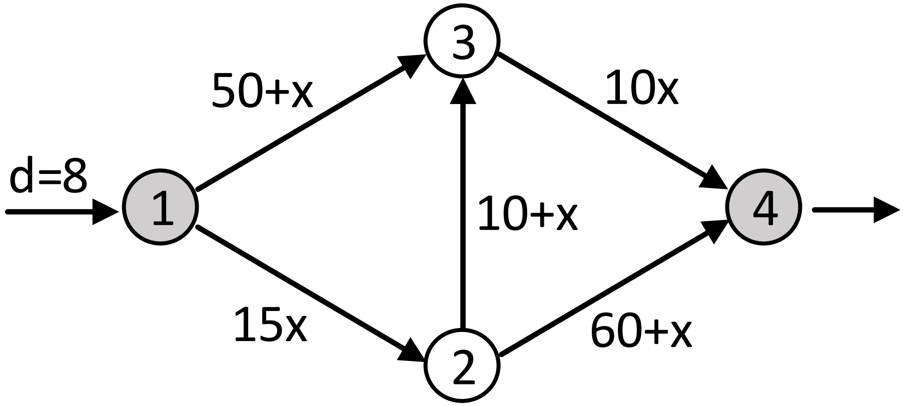

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$$RG = \frac{\mathbf{t \cdot x}}{\mathbf{\kappa \cdot d}} - 1 = \frac{1744}{1040} - 1 = 0.6769$$

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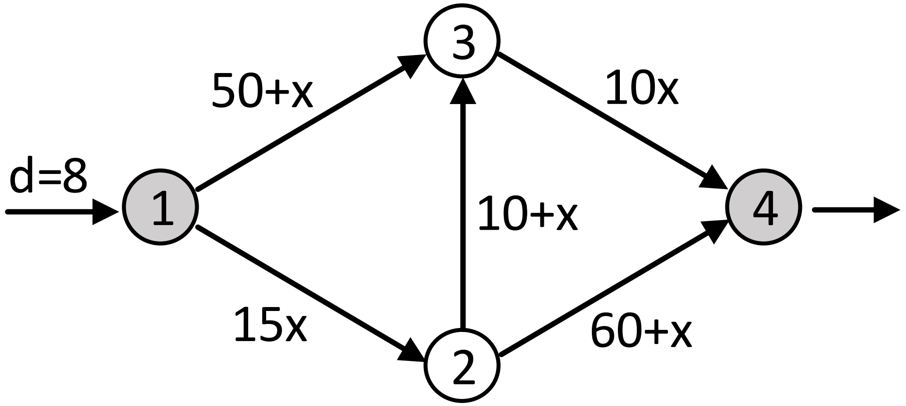

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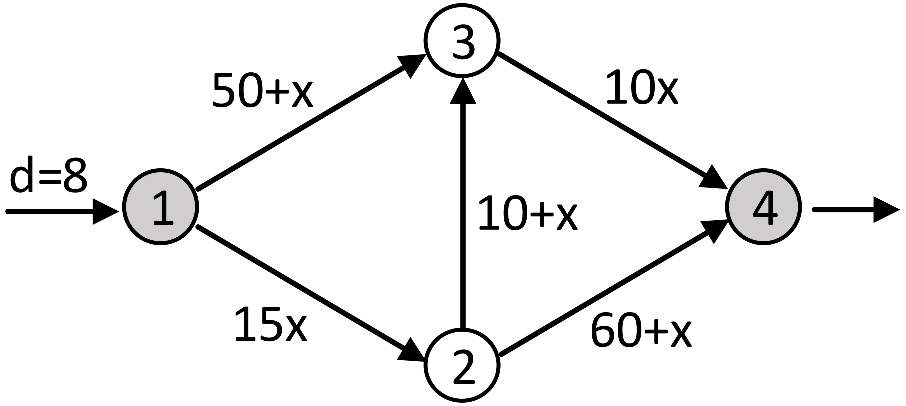
  - [**MSA Iteration $i = 1$**]: Update link flows based on $\mathbf{x} \leftarrow \lambda\mathbf{x}^* + (1-\lambda)\mathbf{x}$
    - MSA step size $\lambda = \frac{1}{i+1} = [\qquad\qquad]$
    - Current link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [\qquad\qquad\qquad\qquad]$ ← From the previous iteration
    - Perform AON assignment based on the current solution: AON path flows $\mathbf{h} = [h^1, h^2, h^3]$ = [ ]
    - Target link flows $\mathbf{x}^* = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [\qquad\qquad\qquad\qquad]$ ← From AON path flows

Exercise



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    - link travel times $\mathbf{t} = [t_{12}, t_{13}, t_{23}, t_{24}, t_{34}] = [120,50,18,60,80]$
  - Find the shortest path using the current solution
    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 120 + 60 = 180$
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 50 + 80 = 130$ ← **SHORTEST PATH**
    - Cost of path $\pi_3 : c^3 = t_{12} + t_{23} + t_{34} = 120 + 18 + 80 = 218$

Check Convergence

$$RG = \frac{\mathbf{t}\cdot\mathbf{x}}{\boldsymbol{\kappa}\cdot\mathbf{d}} - 1 = \frac{1744}{1040} - 1 = 0.6769$$

$$AEC = \frac{\mathbf{t}\cdot\mathbf{x} - \boldsymbol{\kappa}\cdot\mathbf{d}}{\mathbf{d}\cdot\mathbf{1}} = \frac{1744-1040}{8} = 88$$

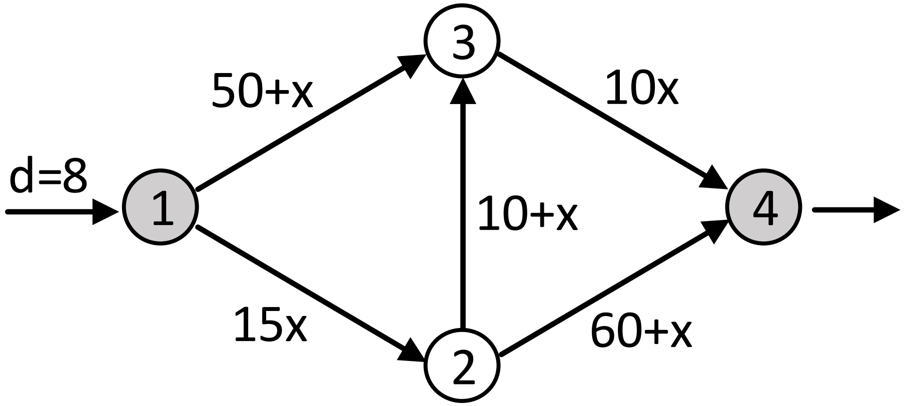
  - [**MSA Iteration $\boldsymbol{i} = 1$**]: Update link flows based on $\mathbf{x} \leftarrow \lambda\mathbf{x}^* + (1-\lambda)\mathbf{x}$
    - MSA step size $\lambda = \frac{1}{i+1} = \frac{1}{1+1} = \frac{1}{2} = 0.5$
    - Current link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [8,0,8,0,8]$ ← From the previous iteration
    - Perform AON assignment based on the current solution: AON path flows $\mathbf{h} = [h^1, h^2, h^3] = [0, 8, 0]$
    - Target link flows $\mathbf{x}^* = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [0,8,0,0,8]$ ← From AON path flows

Compute the new link flows, $\mathbf{x}$

$$\mathbf{x} \quad = \quad \lambda\ \mathbf{x}^* + \quad (1-\lambda)\ \mathbf{x}$$

$$\begin{bmatrix} ? \\ ? \\ ? \\ ? \\ ? \end{bmatrix} = 0.5 \begin{bmatrix} 0 \\ 8 \\ 0 \\ 0 \\ 8 \end{bmatrix} + (1-0.5) \begin{bmatrix} 8 \\ 0 \\ 8 \\ 0 \\ 8 \end{bmatrix}$$

Exercise



Consider the network on the right with one OD-pair from node 1 to node 4 and OD demand of 8 vehicles.

- **Perform the iterative UE Assignment using the MSA method.**
  - Paths: $\pi_1 = [1,2,4],\ \pi_2 = [1,3,4],\ \pi_3 = [1,2,3,4]$
  - [**Iteration 0**]: **The current solution**
    - link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [8,0,8,0,8]$
    - link travel times $\mathbf{t} = [t_{12}, t_{13}, t_{23}, t_{24}, t_{34}] = [120,50,18,60,80]$
  - Find the shortest path using the current solution
    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 120 + 60 = 180$
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 50 + 80 = 130$ ← **SHORTEST PATH**
    - Cost of path $\pi_3 : c^3 = t_{12} + t_{23} + t_{34} = 120 + 18 + 80 = 218$

Check Convergence

$$RG = \frac{\mathbf{t}\cdot\mathbf{x}}{\boldsymbol{\kappa}\cdot\mathbf{d}} - 1 = \frac{1744}{1040} - 1 = 0.6769$$

$$AEC = \frac{\mathbf{t}\cdot\mathbf{x} - \boldsymbol{\kappa}\cdot\mathbf{d}}{\mathbf{d}\cdot\mathbf{1}} = \frac{1744-1040}{8} = 88$$

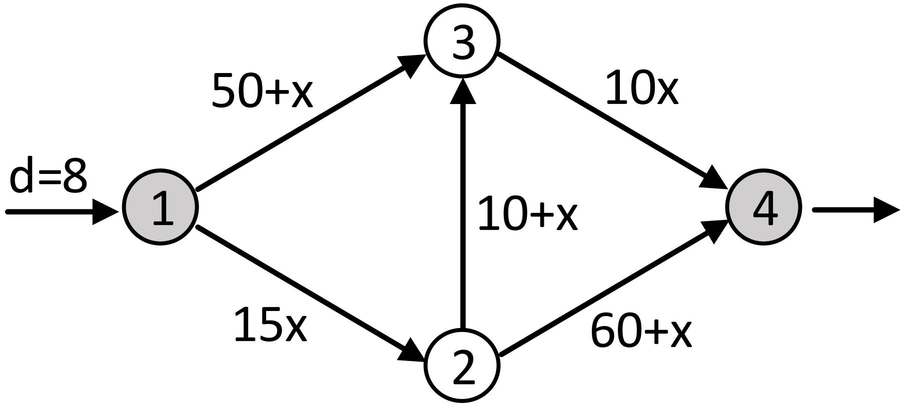
- [**MSA Iteration $\boldsymbol{i} = 1$**]: Update link flows based on $\mathbf{x} \leftarrow \lambda\mathbf{x}^* + (1-\lambda)\mathbf{x}$
  - MSA step size $\lambda = \frac{1}{i+1} = \frac{1}{1+1} = \frac{1}{2} = 0.5$
  - Current link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [8,0,8,0,8]$ ← From the previous iteration
  - Perform AON assignment based on the current solution: AON path flows $\mathbf{h} = [h^1, h^2, h^3] = [0, 8, 0]$
  - Target link flows $\mathbf{x}^* = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [0,8,0,0,8]$ ← From AON path flows

Compute the new link flows, $\mathbf{x}$

$$\mathbf{x} = \lambda\ \mathbf{x}^* + (1-\lambda)\ \mathbf{x}$$

$$\begin{bmatrix} 4 \\ 4 \\ 4 \\ 0 \\ 8 \end{bmatrix} = 0.5 \begin{bmatrix} 0 \\ 8 \\ 0 \\ 0 \\ 8 \end{bmatrix} + (1-0.5) \begin{bmatrix} 8 \\ 0 \\ 8 \\ 0 \\ 8 \end{bmatrix}$$

Exercise



Consider the network on the right with one OD-pair from node 1 to node 4 and OD demand of 8 vehicles.

- **Perform the iterative UE Assignment using the MSA method.**
  - Paths: $\pi_1 = [1,2,4],\ \pi_2 = [1,3,4],\ \pi_3 = [1,2,3,4]$
  - [**MSA Iteration 1**]: **The current solution**
    - link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [4,4,4,0,8]$
    - link travel times $\mathbf{t} = [t_{12}, t_{13}, t_{23}, t_{24}, t_{34}] = [60,54,14,60,80]$
  - Find the shortest path using the current solution
    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 60 + 60 = 120$ ← **SHORTEST PATH**
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 54 + 80 = 134$
    - Cost of path $\pi_3 : c^3 = t_{12} + t_{23} + t_{34} = 60 + 14 + 80 = 154$

Check Convergence

Previously:

$$RG = \frac{\mathbf{t} \cdot \mathbf{x}}{\boldsymbol{\kappa} \cdot \mathbf{d}} - 1 = \frac{1744}{1040} - 1 = 0.6769$$

$$AEC = \frac{\mathbf{t} \cdot \mathbf{x} - \boldsymbol{\kappa} \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{1}} = \frac{1744 - 1040}{8} = 88$$

Currently:

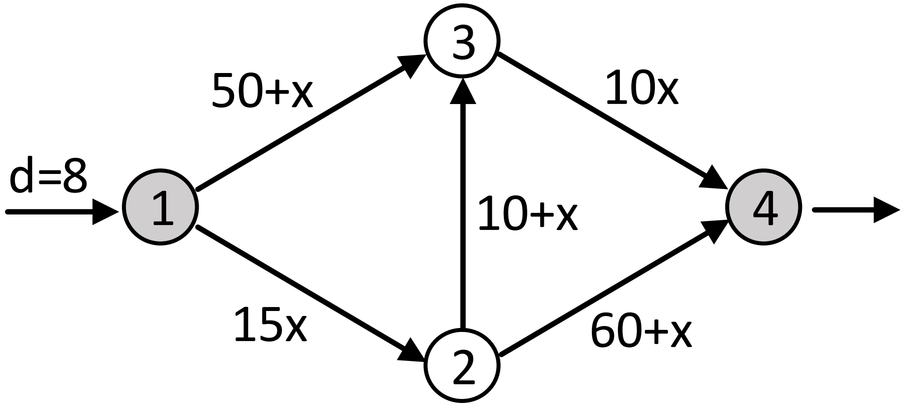
$\mathbf{t} \cdot \mathbf{x}$ =60(4)+54(4)+14(4)+60(0)+80(8)=1152

$\boldsymbol{\kappa} \cdot \mathbf{d}$ =120(8)=960

$$\gamma = \frac{\mathbf{t} \cdot \mathbf{x}}{\boldsymbol{\kappa} \cdot \mathbf{d}} - 1 = \frac{1152}{960} - 1 = 1.2 - 1 = 0.2$$

$$AEC = \frac{\mathbf{t} \cdot \mathbf{x} - \boldsymbol{\kappa} \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{1}} = \frac{1152 - 960}{8} = \frac{192}{8} = 24$$

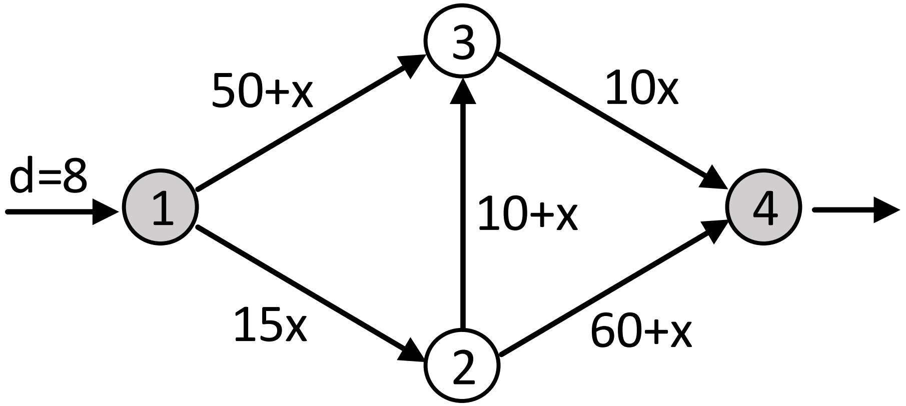
Exercise



Consider the network on the right with one OD-pair from node 1 to node 4 and OD demand of 8 vehicles.

- **Perform the iterative UE Assignment using the MSA method.**
  - Paths: $\pi_1 = [1,2,4]$, $\pi_2 = [1,3,4]$, $\pi_3 = [1,2,3,4]$
  - [**MSA Iteration 1**]: **The current solution**
    - link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [4,4,4,0,8]$
    - link travel times $\mathbf{t} = [t_{12}, t_{13}, t_{23}, t_{24}, t_{34}] = [60,54,14,60,80]$
  - Find the shortest path using the current solution
    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 60 + 60 = 120$ ← **SHORTEST PATH**
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 54 + 80 = 134$
    - Cost of path $\pi_3 : c^3 = t_{12} + t_{23} + t_{34} = 60 + 14 + 80 = 154$
  - [**MSA Iteration $\boldsymbol{i} = \mathbf{2}$**]: Update link flows based on $\mathbf{x} \leftarrow \lambda\mathbf{x}^* + (1-\lambda)\mathbf{x}$
    - MSA step size $\lambda = \frac{1}{i+1}$ = [ ]
    - Current link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}]$ = [ ]← From the previous iteration
    - Perform AON assignment based on the current solution: AON path flows $\mathbf{h} = [h^1, h^2, h^3]$ = [ ]
    - Target link flows $\mathbf{x}^* = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}]$ = [ ]← From AON path flows

Exercise



Consider the network on the right with one OD-pair from node 1 to node 4 and OD demand of 8 vehicles.

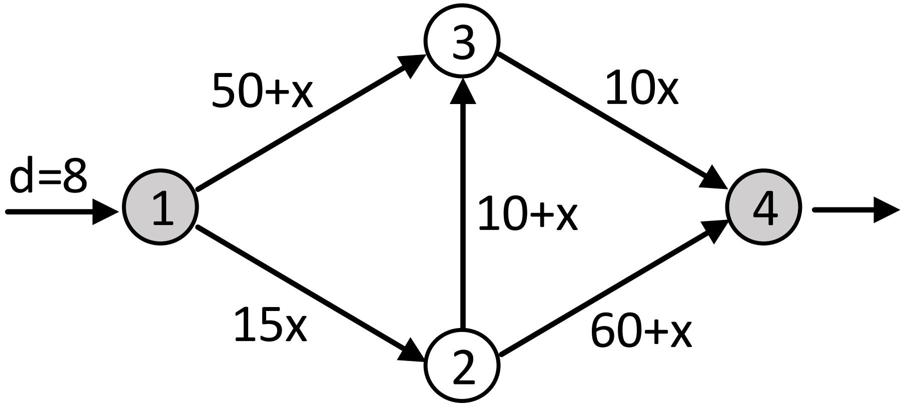
- **Perform the iterative UE Assignment using the MSA method.**
  - Paths: $\pi_1 = [1,2,4]$, $\pi_2 = [1,3,4]$, $\pi_3 = [1,2,3,4]$
  - [**MSA Iteration 1**]: **The current solution**
    - link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [4,4,4,0,8]$
    - link travel times $\mathbf{t} = [t_{12}, t_{13}, t_{23}, t_{24}, t_{34}] = [60,54,14,60,80]$
  - Find the shortest path using the current solution
    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 60 + 60 = 120$ ← **SHORTEST PATH**
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 54 + 80 = 134$
    - Cost of path $\pi_3 : c^3 = t_{12} + t_{23} + t_{34} = 60 + 14 + 80 = 154$
  - [**MSA Iteration $\boldsymbol{i} = \mathbf{2}$**]: Update link flows based on $\mathbf{x} \leftarrow \lambda \mathbf{x}^* + (1 - \lambda)\mathbf{x}$
    - MSA step size $\lambda = \frac{1}{i+1} = \frac{1}{2+1} = \frac{1}{3} = 0.3333333$
    - Current link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [4,4,4,0,8]$ ← From the previous iteration
    - Perform AON assignment based on the current solution: AON path flows $\mathbf{h} = [h^1, h^2, h^3] = [8, 0, 0]$
    - Target link flows $\mathbf{x}^* = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [8,0,0,8,0]$ ← From AON path flows

Compute the new link flows, $\mathbf{x}$

$$\mathbf{x} = \lambda\ \mathbf{x}^* + (1-\lambda)\ \mathbf{x}$$

$$\begin{bmatrix} ? \\ ? \\ ? \\ ? \\ ? \end{bmatrix} = 0.333 \begin{bmatrix} 8 \\ 0 \\ 0 \\ 8 \\ 0 \end{bmatrix} + (1 - 0.333) \begin{bmatrix} 4 \\ 4 \\ 4 \\ 0 \\ 8 \end{bmatrix}$$

Exercise



Consider the network on the right with one OD-pair from node 1 to node 4 and OD demand of 8 vehicles.

- **Perform the iterative UE Assignment using the MSA method.**
  - Paths: $\pi_1 = [1,2,4],\ \pi_2 = [1,3,4],\ \pi_3 = [1,2,3,4]$
  - [**MSA Iteration 1**]: **The current solution**
    - link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [4,4,4,0,8]$
    - link travel times $\mathbf{t} = [t_{12}, t_{13}, t_{23}, t_{24}, t_{34}] = [60,54,14,60,80]$
  - Find the shortest path using the current solution
    - Cost of path $\pi_1 : c^1 = t_{12} + t_{24} = 60 + 60 = 120$ ← **SHORTEST PATH**
    - Cost of path $\pi_2 : c^2 = t_{13} + t_{34} = 54 + 80 = 134$
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  - [**MSA Iteration $\boldsymbol{i = 2}$**]: Update link flows based on $\mathbf{x} \leftarrow \lambda \mathbf{x}^* + (1 - \lambda)\mathbf{x}$
    - MSA step size $\lambda = \frac{1}{i+1} = \frac{1}{2+1} = \frac{1}{3} = 0.3333333$
    - Current link flows $\mathbf{x} = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [4,4,4,0,8]$ ← From the previous iteration
    - Perform AON assignment based on the current solution: AON path flows $\mathbf{h} = [h^1, h^2, h^3] = [8, 0, 0]$
    - Target link flows $\mathbf{x}^* = [x_{12}, x_{13}, x_{23}, x_{24}, x_{34}] = [8,0,0,8,0]$ ← From AON path flows

Compute the new link flows, $\mathbf{x}$

$$\mathbf{x} = \lambda\,\mathbf{x}^* + (1-\lambda)\,\mathbf{x}$$

$$\begin{bmatrix} 5.333 \\ 2.667 \\ 2.667 \\ 2.667 \\ 5.333 \end{bmatrix} = 0.333 \begin{bmatrix} 8 \\ 0 \\ 0 \\ 8 \\ 0 \end{bmatrix} + 0.667 \begin{bmatrix} 4 \\ 4 \\ 4 \\ 0 \\ 8 \end{bmatrix}$$

Pseudo Code for Traffic Assignment using MSA method

- An iterative process that updates link flows ($\mathbf{x}$) using $\mathbf{x} \leftarrow \lambda\mathbf{x}^* + (1-\lambda)\mathbf{x}$ at each iteration by identifying target link flows ($\mathbf{x}^*$) and step size (λ):
  - Target link flows ($\mathbf{x}^*$) from AON assignment
  - Step size (λ) from MSA method

```
max_iter = 10000        Set the maximum iteration number to avoid a potential infinite loop
target_rg = 0.0001      Target threshold value for the relative gap (RG)
target_aec = 0.0001     Target threshold value for the average excess cost (AEC)

Initialisation:
Set initial link costs to free-flow travel times
Perform All-or-Nothing (AON) assignment to get initial link flows
Update link costs

step_size = 1
iteration = 0

while iteration <= max_iter:

    Find the shortest path and time (κ^{rs}) for each OD

    Calculate the convergence measures (RG and AEC).

    If RG < target_rg and AEC < target_aec,     Stop if target convergence was achieved
        stop

    If iteration = max_iter,     Stop if the maximum allowed iteration number was reached
        stop

    iteration += 1     Continue to the next iteration by increasing the iteration number by one

    Perform All-or-Nothing assignment to get target link flows

    step_size <- Assign the MSA step size based on the current iteration number

    Shift link flows by updating x ← λx* + (1 − λ)x, where λ is step_size

    Update link costs
```

Thank you